1. Two Sum

Easy | Algorithms

Given an array of integers `nums` and an integer `target` , return *indices* of the two numbers such that they add up to target.

You may assume that each input would have **exactly one solution**, and you may not use the same element twice.

You can return the answer in any order.

Examples

Example 1:

```
Input: nums = [2,7,11,15], target = 9
Output: [0,1]
```

Explanation: Because nums[0] + nums[1] == 9 , we return [0, 1] .

Example 2:

```
Input: nums = [3,2,4], target = 6
Output: [1,2]
```

Example 3:

```
Input: nums = [3,3], target = 6
Output: [0,1]
```

Constraints

- 2 <= nums.length <= 10^4
- -10^9 <= nums[i] <= 10^9
- -10^9 <= target <= 10^9
- Only one valid answer exists.

Follow-ups

- Can you come up with an algorithm that is less than $O(n^2)$ time complexity?